

The empirical recurrence for OEIS A316637, recovered from its tail

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Abstract

OEIS A316637 records “Empirical recurrence of order 69” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 71$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 72$. Its last coefficient is $c_{69} = 64 \neq 0$, so no further tail satisfies anything shorter; any order-69 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A316637 is “Number of $n \times 4$ 0..1 arrays with every element unequal to 1, 2, 3, 5, 7 or 8 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

2, 45, 122, 524, 2342, 9773, 44095, 195387, ...

The entry states:

Empirical recurrence of order 69 (see link above)

As of the “Last modified” line on the live entry (Jul 09 2018 11:17:18, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 71$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 71$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 69: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 69.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 142$ exact terms from $n = 3$ on, it returns order 69 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{69} c_i a(n-i),$$

c_1	3	c_{19}	−159187	c_{37}	19061436	c_{55}	54931
c_2	7	c_{20}	92180	c_{38}	20460052	c_{56}	400761
c_3	26	c_{21}	391295	c_{39}	20156923	c_{57}	−289745
c_4	−87	c_{22}	867931	c_{40}	16293342	c_{58}	−59161
c_5	−143	c_{23}	1748187	c_{41}	9706689	c_{59}	−29579
c_6	−364	c_{24}	1764169	c_{42}	5969997	c_{60}	−85240
c_7	613	c_{25}	1807830	c_{43}	1463264	c_{61}	5162
c_8	1140	c_{26}	19008	c_{44}	597946	c_{62}	−10982
c_9	2707	c_{27}	−3393101	c_{45}	964117	c_{63}	−15724
c_{10}	−192	c_{28}	−6255507	c_{46}	−2631351	c_{64}	−1720
c_{11}	−1653	c_{29}	−10108644	c_{47}	−3028118	c_{65}	−252
c_{12}	−7438	c_{30}	−11695773	c_{48}	−2077782	c_{66}	232
c_{13}	−3801	c_{31}	−11716225	c_{49}	−2845211	c_{67}	888
c_{14}	−7572	c_{32}	−8876437	c_{50}	217198	c_{68}	432
c_{15}	−3520	c_{33}	−2347736	c_{51}	21746	c_{69}	64
c_{16}	−11078	c_{34}	4278304	c_{52}	−1475224		
c_{17}	−71146	c_{35}	10934384	c_{53}	254954		
c_{18}	−81227	c_{36}	15430851	c_{54}	−323056		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 72$, and it is the recurrence OEIS A316637 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{69} - c_1 x^{69-1} - \dots - c_{69}$. Its constant term is $-c_{69} = -64$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the

minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 69 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 69, so $\deg q = 69$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 69, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 22 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 69, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 64.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-69 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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